

Heaven's light is our guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY  
Department of Mechatronics Engineering

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## Lab Report

***Experiment no.:08***

***Experiment name: Designing and Solving Real life problems using PLC  
programming***

***Submitted By***

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**MTE 4102**

**Automation Sessional**

## **Experiment no.:08**

### **Experiment name: Designing and Solving Real life problems using PLC programming**

#### **Objectives:**

- To apply PLC programming using OpenPLC software in solving real-life automation problems.
- To understand the use of timers, counters, and sequential logic in practical applications.
- To gain hands-on experience in designing safe, reliable, and efficient control systems.

#### **Theory:**

Programmable Logic Controllers (PLCs) are specialized digital controllers used for industrial and automation purposes. They execute logical instructions based on inputs and generate outputs to control machines or processes. Key functions include timers for time-based operations, counters for event tracking, and sequential logic for step-by-step control. Additionally, safety mechanisms such as alarms and interlocks are vital in preventing accidents and ensuring smooth operation. OpenPLC software provides a platform to simulate and test these logic-based systems before real-world implementation, making it an effective learning tool for understanding automation.

#### **Required Apparatus:**

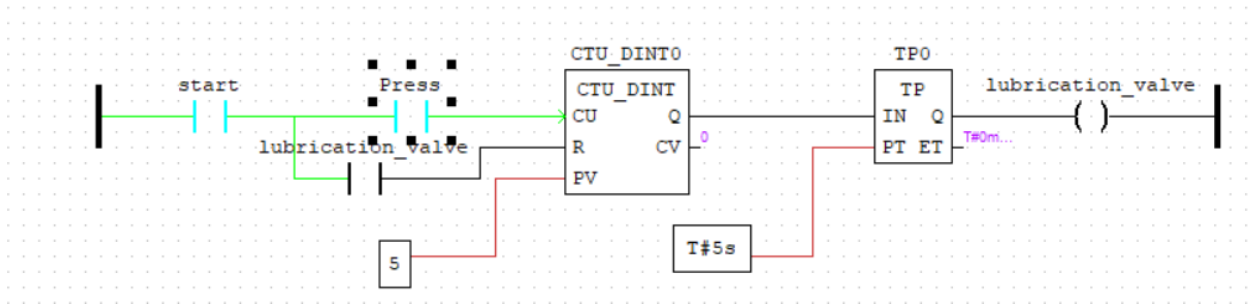
1. Laptop
2. OpenPLC Editor (PC Software)

## Problem Statements & Ladder Diagram:

### Problem 1:

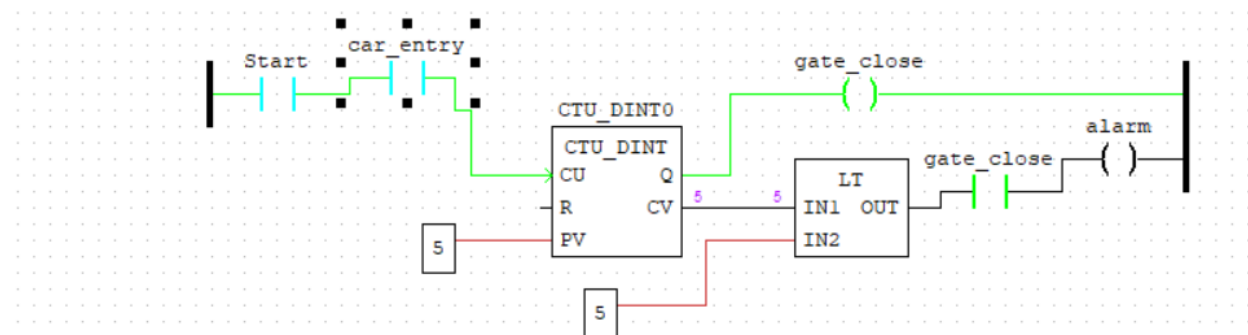
A stamping press turns on lubrication valve after 5 press, stays on for 5 s and then repeats the action.

### Ladder Diagram:

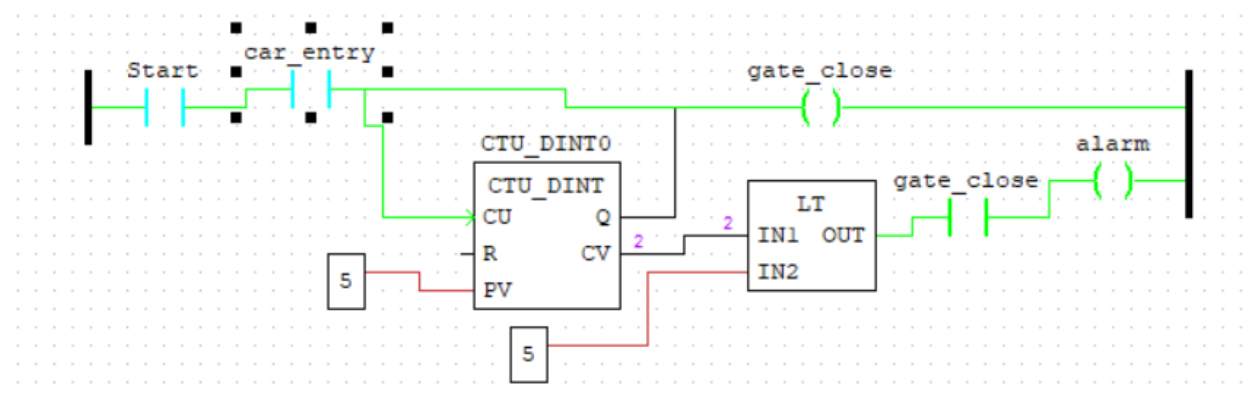


**Problem 2:** A parking lot has a capacity for 5 cars , When 5 cars enter the gate is off , if the gate doesn't close after 5 cars have entered, the alarm rings.

### Ladder Diagram:

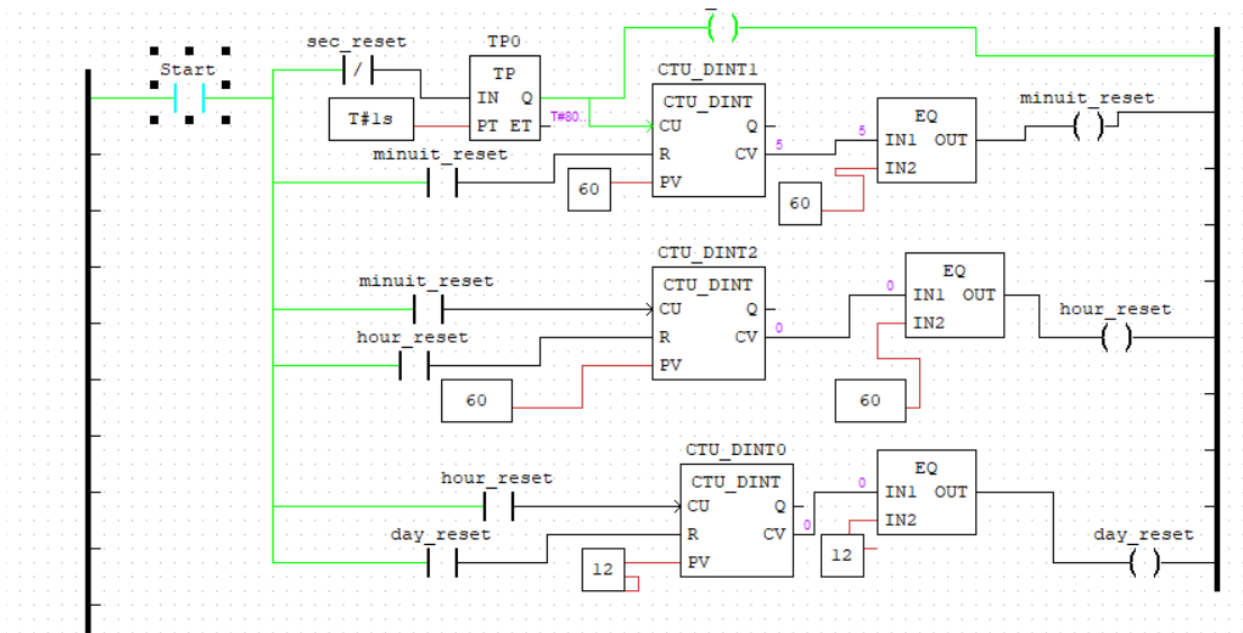


When the number of car is not 5, If the gate is closed forcefully , the alarm will Turns on.



### Problem 3:

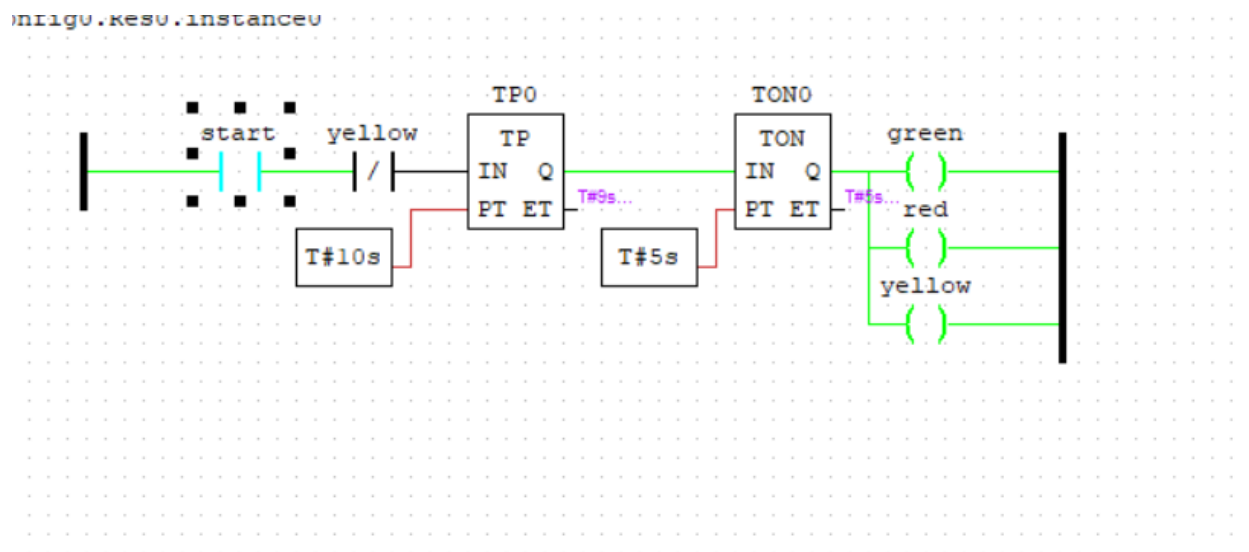
Design of a digital clock with display



### Problem 4:

Green, yellow, red lights simultaneously on repeats with 5s interval

Ladder Diagram:



## **Discussion:**

In this lab, four real-life problems were solved using PLC programming to demonstrate different aspects of automation. The stamping press lubrication system applied counters and timers to ensure the valve activated automatically after every five presses, highlighting preventive machine maintenance. The parking lot problem introduced both counting and safety features, where the gate closed after five cars entered and triggered an alarm if it failed to close, reflecting real-world interlocks. The digital clock task used timers and display logic to count seconds, minutes, and hours, showing how precise sequencing is possible through PLCs. Finally, the traffic light system applied sequential logic with fixed intervals for green, yellow, and red lights, simulating traffic management systems. Together, these tasks demonstrated the versatility of PLCs in handling timing, counting, safety, and sequencing, which are essential elements in both industrial and everyday applications.

## **Conclusion:**

The lab successfully demonstrated the application of PLC programming in solving practical automation problems. Through the stamping press, parking lot, digital clock, and traffic light tasks, we learned how to integrate timers, counters, and sequential logic into functional control systems. This enhanced our understanding of automation and showed how PLCs improve efficiency, reliability, and safety in real-world operations.